

PreProd SE: A Governance-Aware Academic Sandbox Platform for Digital Learning and Computational Research

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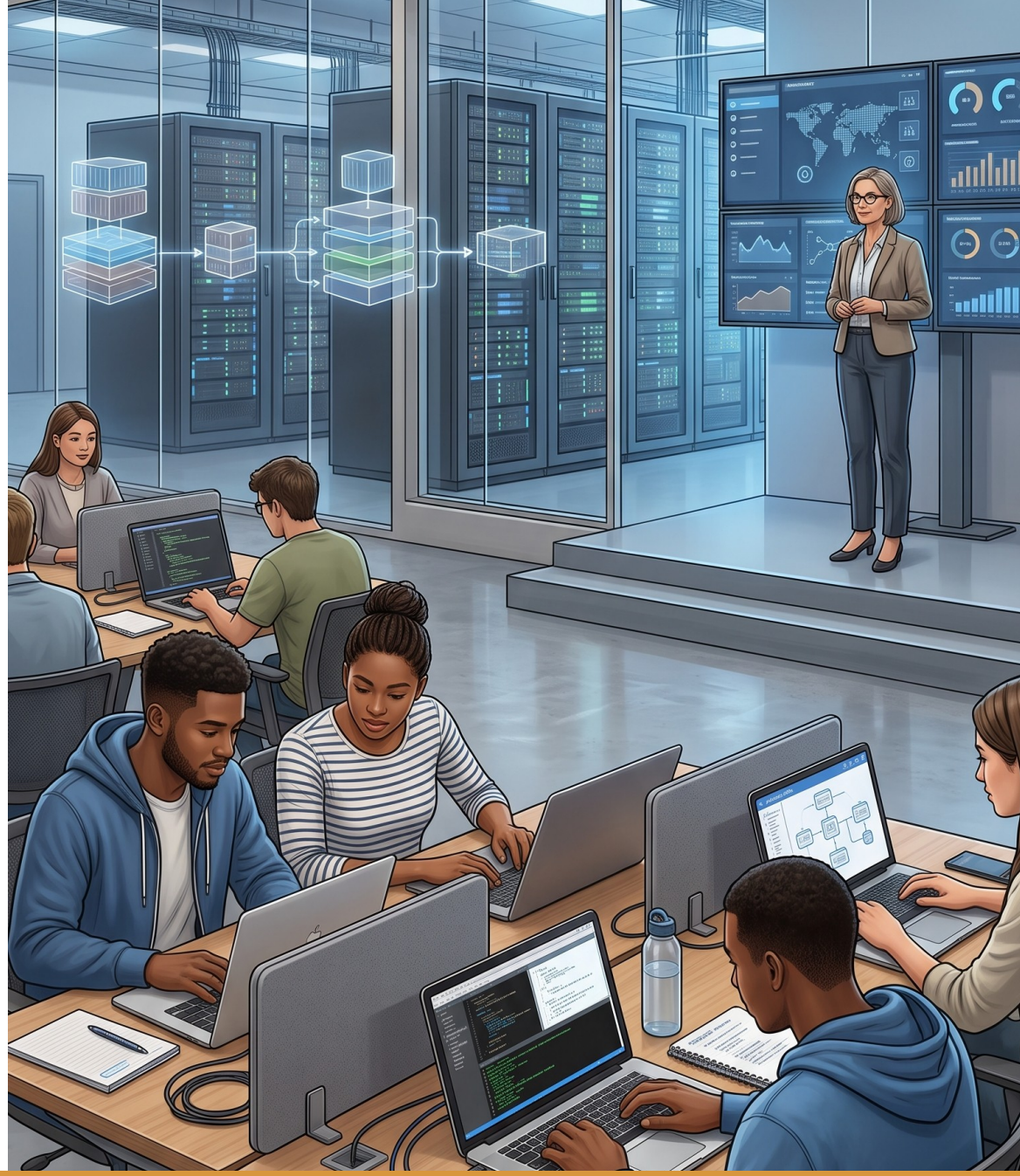
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Introduction

- Practical experience is essential in computing education
- Students must engage in real-world development and deployment
- Universities struggle to provide:
  - Scalable environments
  - Secure systems
  - Supervised learning spaces

Key Issue:

Lack of controlled, deployment-ready academic environments





Limitations of Existing Approaches

Public cloud platforms

- High cost
- No institutional control
- Limited academic supervision

Shared university labs

- Not scalable
- Limited flexibility

Personal devices

- Limited resources
- No standardization

Research Gap

Cloud-native technologies exist ✔

DevOps and CI/CD workflows exist ✔

However:

☐ Not designed for academic environments

☐ Lack supervision and governance

☐ No integration with structured learning

Gap:

No platform combining deployment + governance + education

Proposed Solution?



PreProd SE – Pre-Production Sandbox Enviroment

A governance-aware academic sandbox platform

Proposed Solution

Key Features:

- ✓ Hybrid PaaS + SaaS model
- ✓ Isolated student workspaces
- ✓ Deployment-oriented learning
- ✓ Supervisor-integrated governance



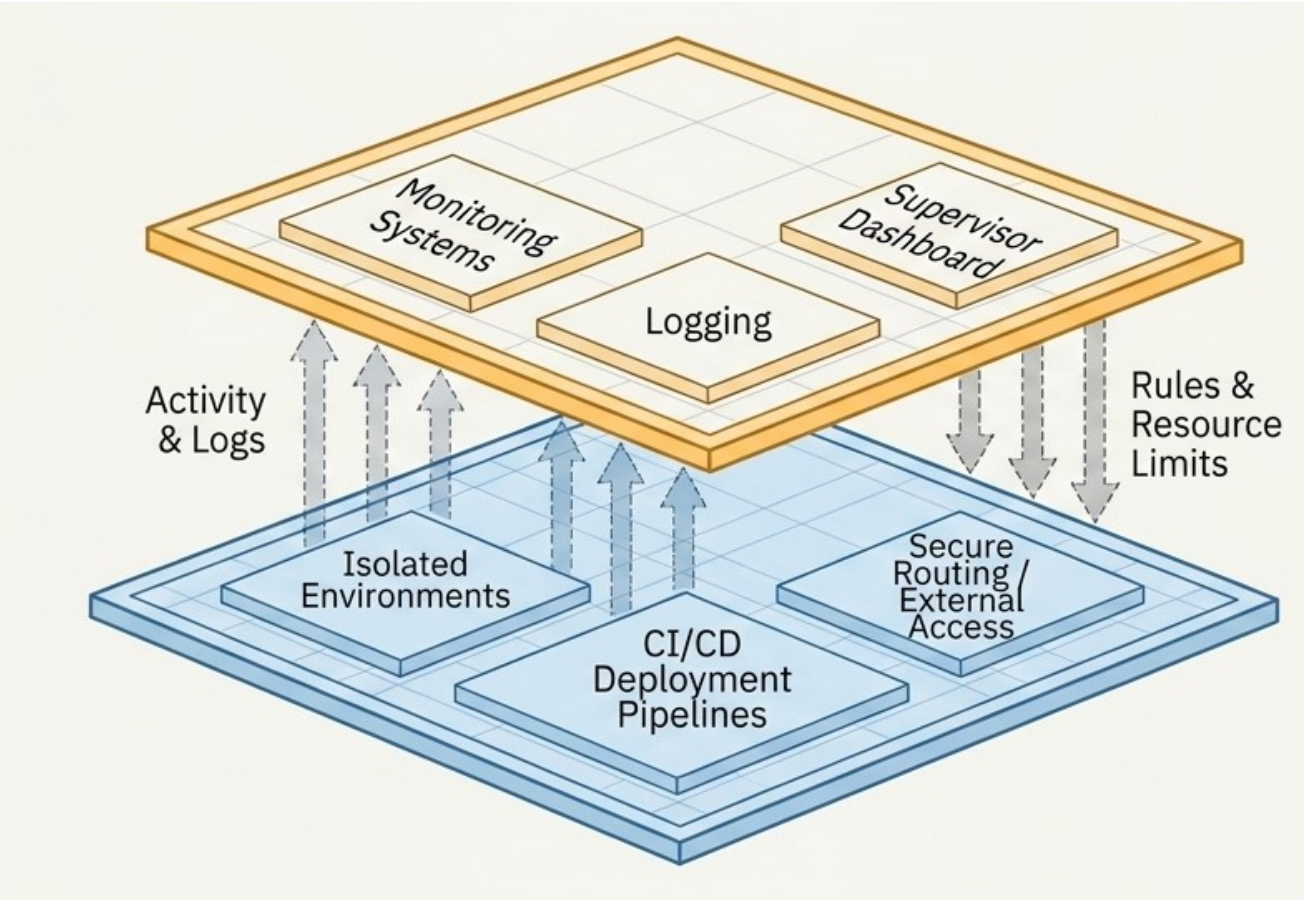
Goal:

A unified framework integrating industry-standard technologies with a governance layer that facilitates supervisor-student interaction, supporting both computational deployments and academic research workflows.

Methodology

How it really works?

System Architecture (Overview)



- ✓ Container-based execution (isolated environments)
- ✓ CI/CD pipelines for automated deployment
- ✓ Secure routing and external access
- ✓ Monitoring and logging systems

Outcome:

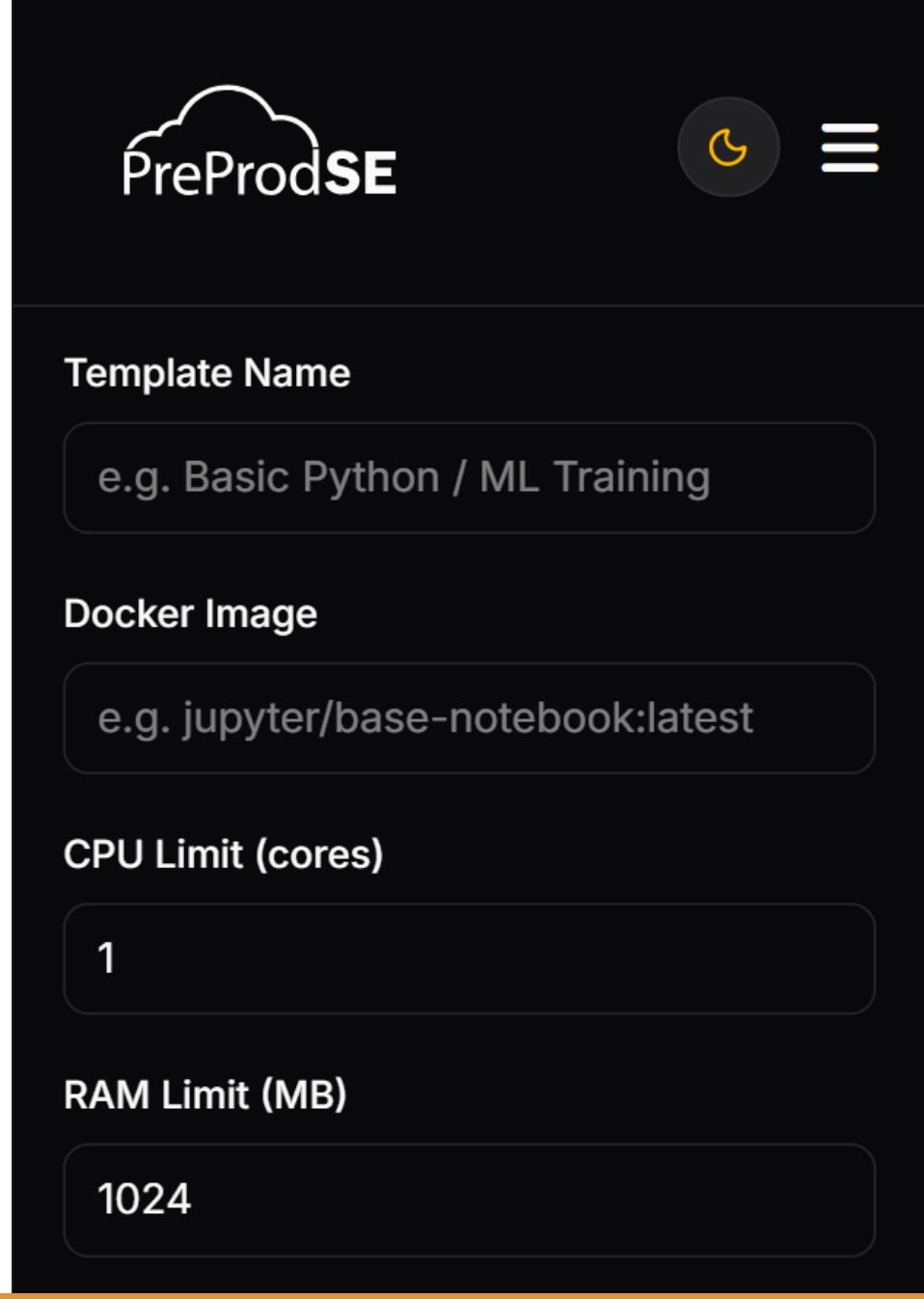
Scalable, flexible, and controlled academic infrastructure

Governance Model (Key Contribution)

- ✓ Predefined resource limits (CPU, memory)
- ✓ Supervisor-controlled resource escalation
- ✓ Monitoring of user activity and deployments
- ✓ Controlled access to workflows
- ✓ Institutional ownership of data and infrastructure

Key Idea:

Balancing the student-supervisor autonomy with academic control



Implementation

Technologies used:

- Docker (containerisation)
- GitHub CI/CD (deployment automation)
- Traefik (routing)
- Cloudflare (secure access)
- Wazuh & Suricata (security monitoring)

Note:

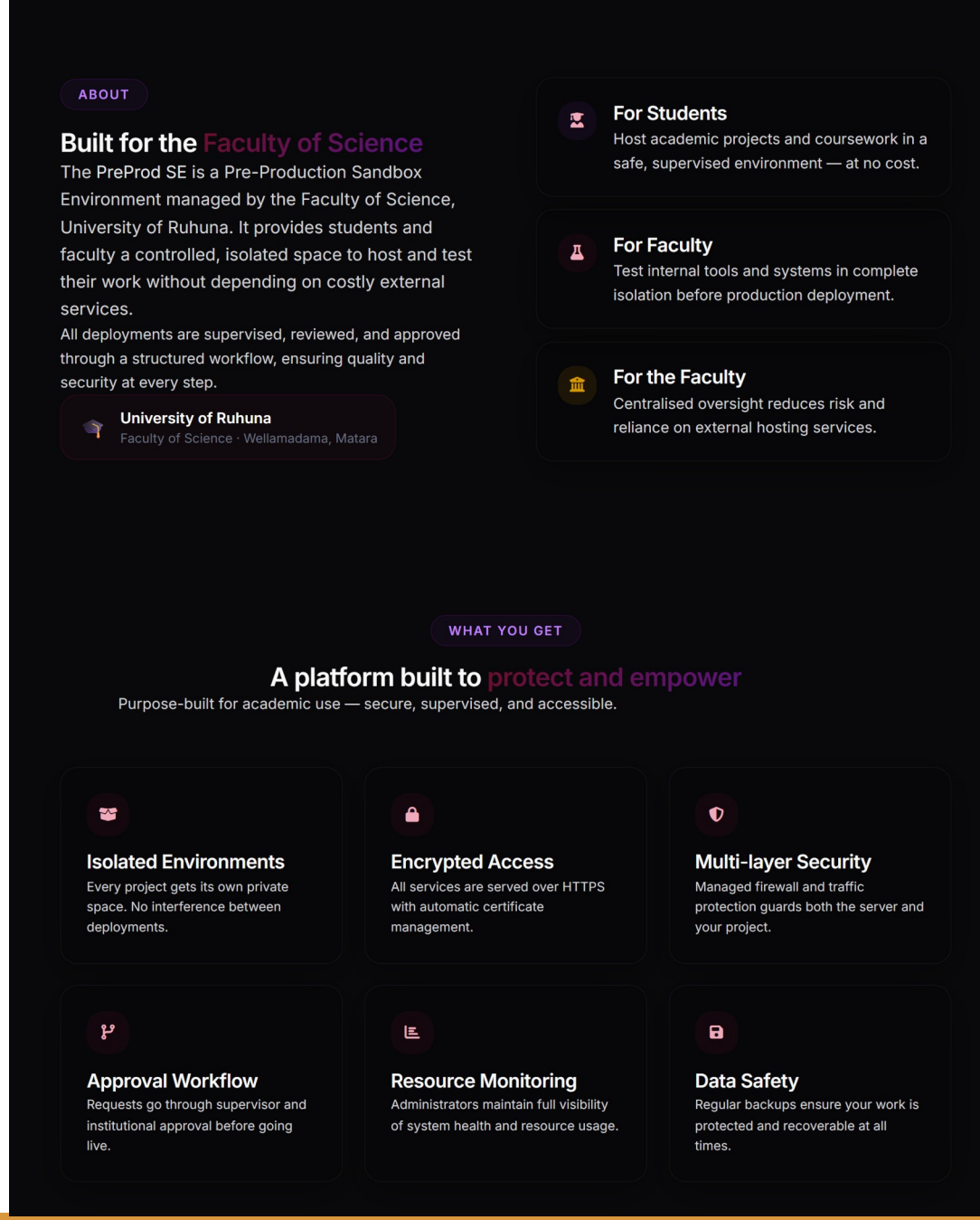
Technologies support the system;
not the main contribution

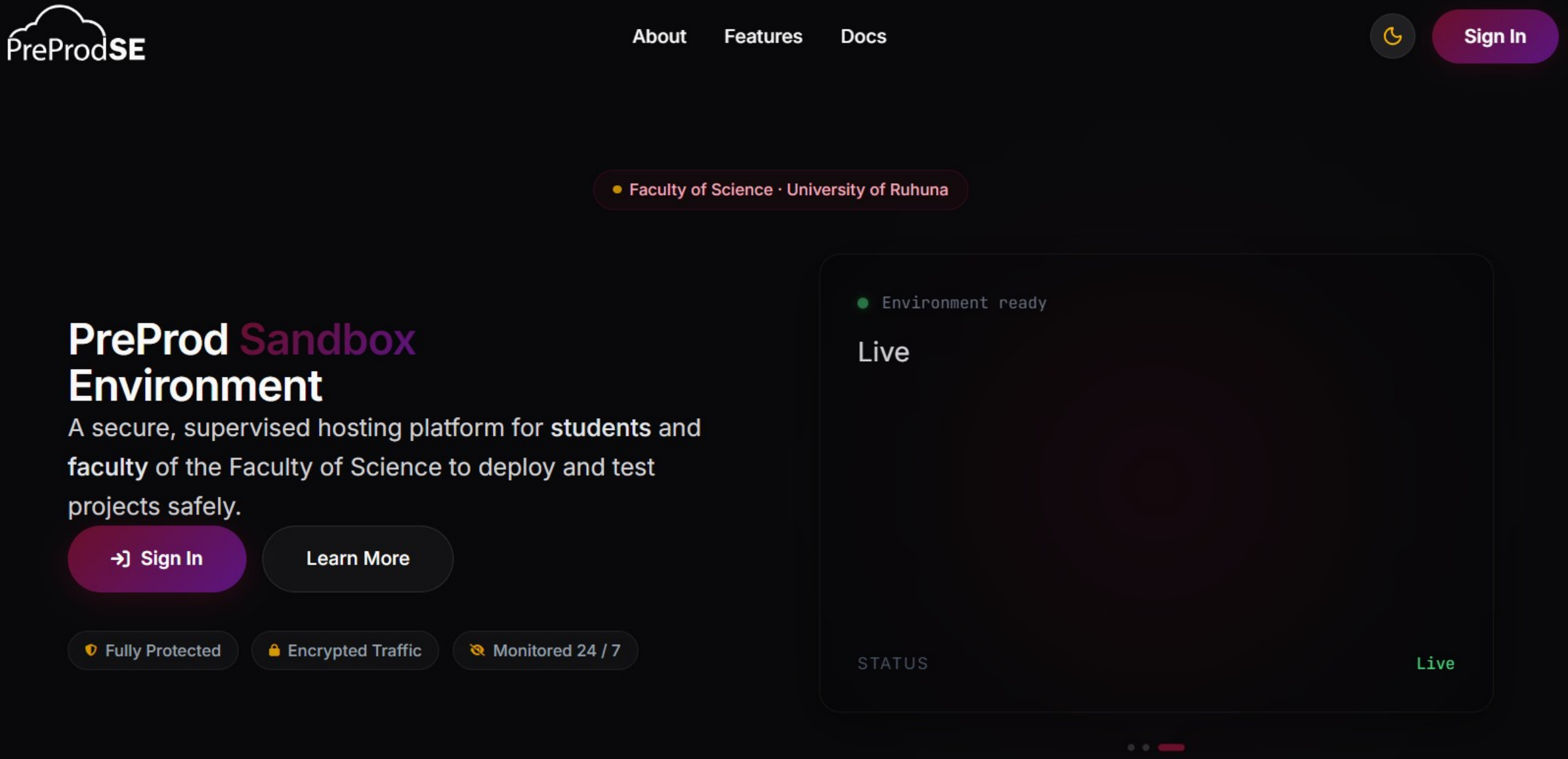
Pilot Deployment & Findings

- PreProd SE v1.0 was launched on **May 2026**
- Conducted in a real academic environment
- 20 students and supervisors

Observed Outcomes:

- ✓ Concurrent isolated workloads
- ✓ No resource conflicts
- ✓ Successful automated deployments
- ✓ Improved supervisory visibility
- ✓ Smooth Resource Handling




PreProdSE
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Features
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PreProd Sandbox Environment
A secure, supervised hosting platform for students and faculty of the Faculty of Science to deploy and test projects safely.
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Fully Protected
Encrypted Traffic
Monitored 24 / 7
Environment ready
Live
STATUS
Live



Contribution & Impact

- ✓ Bridges theory and real-world practice
- ✓ Brought the Student-Supervisor Relationship into the Digital Learning and Computational Research
- ✓ Enables deployment-based learning
- ✓ Integrates governance into digital education

Relevance: Supports digital transformation and governance in education

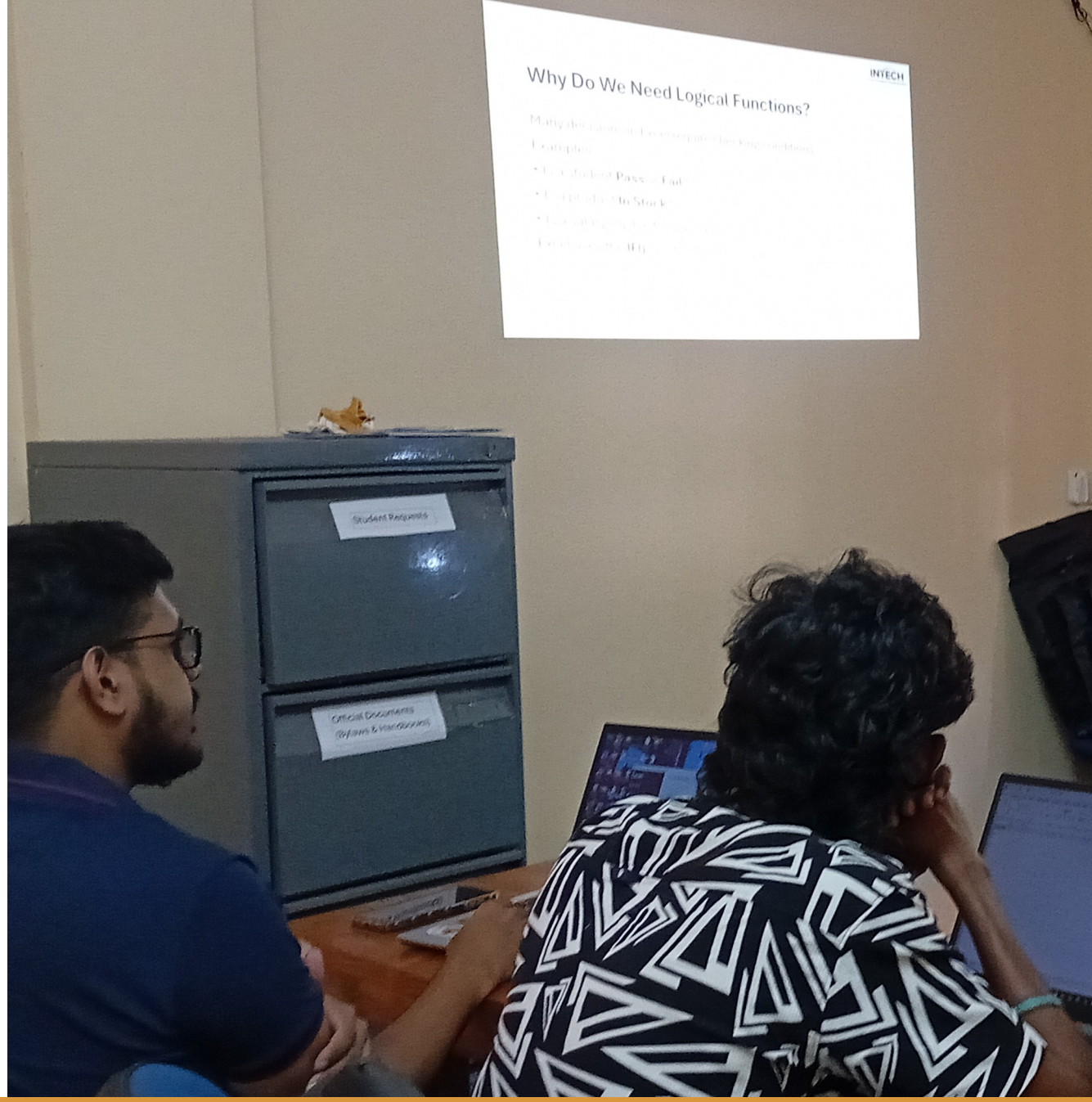
Limitations & Future Work

Limitations:

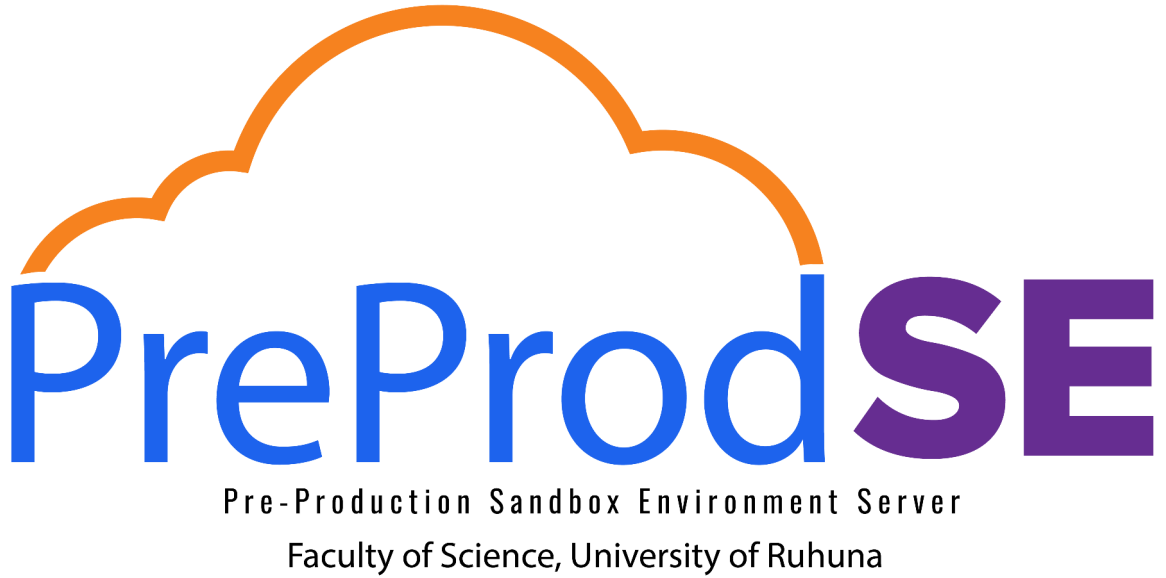
- No formal benchmarking
- Small-scale pilot
- No structured user study

Future Work:

- ✓ Large-scale evaluation
- ✓ Performance analysis
- ✓ Integration with LMS and university systems



Conclusion



- ✓ PreProd SE introduces a governance-aware learning platform
- ✓ Demonstrates feasibility in real academic settings
- ✓ Supports supervised, deployment-oriented education

Final Thought:

Enabling real-world learning within a controlled academic environment

Thank you for your attention.